NODE — Build Brief for Claude Code v1.0

Status: Handoff document for implementation. Derived from design work through 06 Aug 2026.
Audience: Claude Code, working solo on this build. No other coding agents are involved — write and reason as if you are the only implementer, with full continuity across all phases below.
Nature of this document: This is a **stress-testable starting structure, not fixed law.** Every mechanic below has been reasoned through and partially validated by simulation, but nothing is final until it's actually running and being played. Where you find a rule that doesn't hold up once built — breaks, feels wrong, produces degenerate behaviour — say so and propose the fix rather than silently working around it. The design intent (see §0) is the one thing that shouldn't drift; the specific numbers and mechanics implementing it should be treated as hypotheses.

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0. Design Intent — Read This First

NODE is a persistent, multiplayer, social-economic game for a bounded population ("node") of roughly **50–80 players**, replacing warfare-based conflict with social standing, gossip, and economic maneuvering. The engine behind everything is the **"49-51" philosophy**: the system should sit permanently tilted, slight sustained uphill pressure, never balanced, never resolved, never a cliff-edge. Nothing should ever fully collapse (there is always a backstop), but nothing should ever feel comfortable for long either. If a mechanic you're building trends toward comfortable equilibrium or toward catastrophic collapse, it has drifted from the intent — flag it.

Core thesis in one line: **information and social exposure are the game.** There is no combat system. The tension comes from asymmetric information, structural economic pressure, and the fact that you are never fully anonymous and never fully known.

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1. Global State & Notation (carries through every phase below)

Symbol	Meaning
`N`	Node population (total players), target range 50–80, sweet spot **60–65**
`R`	Number of essential role-slots (Baker, Miller, Courier, etc.)
`p_m`	Monthly churn probability per role-holder
`p_d`	Daily churn probability, derived: `p_d = 1 - (1 - p_m)^(1/30)`

Symbol	Meaning
`s_i(t)`	State of role-slot `i` at day `t`: `FILLED`, `VACANT`, or `BACKSTOPPED`
`τ_i(t)`	Days role `i` has been continuously vacant at day `t`
`t_flag`	Day a vacancy becomes publicly visible (~3 days)
`t_hard`	Day a vacancy is force-filled (~14 days, hard backstop)
`γ`	Bertrand price-substitutability coefficient, `γ ∈ [0,1)` in the original spec, but see §2.3 — actual stable range extends further
`p_i(t)`	Price set by role-holder `i` on day `t`
`R(t)`	Player's accrued economic floor balance ("Ruin Floor") — **unresolved, see §7**

Where a parameter is marked **[DERIVED]** it's forced by model structure. Where marked **[CALIBRATED — provisional]** it was chosen to fit simulation runs today, not validated against real playtest data — expect to retune these once real players exist.

PHASE 1 — Economic Core (Miller/Baker Reaction Engine)

Why first: everything else — vacancy pressure, social tension, the Wall's content — sits on top of this. Build and simulate this in isolation before touching UI, identity, or comms. Your own internal simulation harness (see §1.5) should be able to run this headless and produce the same stability curves described below before you connect a single player-facing system to it.

1.1 Structure

Two chained market layers:

- **Millers** — Cournot quantity-competition (upstream). Millers choose output quantity; aggregate supply determines flour price via simple inverse demand.
- **Bakers** — Bertrand price-competition (downstream). Bakers set price given flour cost passed through from Millers; compete on price against each other, not quantity.

This is a real, well-studied market structure (Kreps-Scheinkman two-stage game), not an invented mechanic — implement it as a genuine chained simulation, not a lookup table.

1.2 Miller layer (Cournot)

```
avg_rival_q_i = (sum(q) - q_i) / (n_millers - 1)
q_i(t+1) = 0.5 * q_i(t) + 0.5 * (1 -
avg_rival_q_i) + noise
q_i clipped to [0.01, 1]
total_supply(t) = sum(q_i(t))
```

```
flour_price(t) = clip(1.2 - 0.3 *
total_supply(t), 0.05, 2.0)   [CALIBRATED —
provisional inverse-demand curve]
```

1.3 Baker layer (Bertrand), fed by flour price

```
avg_rival_p_i = (sum(p) - p_i) / (n_bakers - 1)
cost_pressure = flour_price(t) * 0.3
[CALIBRATED — provisional cost passthrough]
p_i(t+1) = (1 - γ/2) * p_i(t) + (γ/2) *
avg_rival_p_i + cost_pressure * 0.1 + noise
p_i clipped to [0, 2]
```

1.4 Validated findings — build these as regression tests, not just design notes

These were confirmed by simulation this session and should be encoded as automated tests once the engine exists, so future changes can't silently break them:

- **The reaction slope `γ/2` is always a contraction mapping** for `γ < 2` — no bifurcation possible in this linear formulation. An earlier draft spec claimed a chaos threshold at `γ < 0.85`; that number does **not** exist in this model and should not be resurrected. The actual boundary found by direct simulation is **`γ = 2`**, not 0.85.
- **Instability is a headcount property, not just a γ property.** At exactly `n=2` players in a role slot, price spread blows up sharply as `γ` approaches/exceeds 2 (spread jumps from ~0.02 to ~0.5). At `n≥3` players in a

slot, the system stays stable well past `γ=2` — it self-averages the shock away. **Design implication: never let a role slot sit at exactly 2 players for any length of time** — this is a hard input into the vacancy backstop logic in Phase 2, not just a balance nicety.

- **Miller headcount, not Baker headcount, drives system volatility.** More Millers → lower flour price + higher baker-side volatility (more upstream Cournot jostling propagates downstream). Baker headcount (3, 4, or 5) barely changes outcomes once flour price is set. **Design implication: keep Miller count deliberately low (2–3) as the structurally scarce upstream role** — this is the real tension lever, not symmetric Miller:Baker ratios.
- Miller scarcity is **exposure, not privilege** — structural power, structural workload, structural target, self-balancing by player temperament rather than a game-enforced reward. Do not build any mechanic that pays Millers a structural bonus for holding the role. The pressure itself is the entire design payload.

1.5 Recommended role-slot mix at N=60–65

Given ~1/3 of players hold role-slots (~21 people) and ~2/3 are pure social/gossip-layer players (no essential role, see Phase 4 coda):

- 2 "thin" rivalry roles (e.g. Miller) at **3 players each** — deliberately just above the dangerous n=2 cliff, maximally tense but structurally safe.

- 4 "stable" roles (e.g. Baker, Courier, etc.) at **3–4 players each** — all slots ≥3.

Build a headless simulation harness that can sweep `N`, `R`, `γ`, and per-role headcounts and output stability curves — you'll want this for Phase 2 and Phase 6 regardless.

PHASE 2 — Vacancy, Churn & the Backstop System

Why second: this system consumes the economic core's role-slot state directly and is what converts "a player quit" into visible social/economic pressure — the connective tissue between the economy and the social layer that comes later.

2.1 Vacancy as a semi-Markov process

```
FILLED --(p_d)--> VACANT --(λ_fill(τ) or
τ≥t_hard)--> FILLED
```

2.2 Vacancy fill hazard function

```
λ_fill(τ) = 1 - (1 - p_c(τ))^(N - R)

p_c(τ) = β · (0.2 + 0.8 · min(τ / T_pain, 1)) ·
V(τ)
```

- `β` **[CALIBRATED — provisional, = 0.0008]** — baseline willingness-per-candidate-per-day.

- `T_pain` **[CALIBRATED — provisional, = 14 days]** — time for economic pressure to reach plateau. Motivated by (not derived from) Mullainathan & Shafir's scarcity/bandwidth-tax framework — pressure should build, not appear instantaneously.
- `V(τ)` — visibility multiplier:

  ```
  V(τ) = 1        if τ < t_flag
  V(τ) = v_boost  if τ ≥ t_flag
  ```

  `v_boost` **[CALIBRATED — provisional, = 3.0]** — models the vacancy becoming publicly/socially visible at `t_flag`.

2.3 Two-stage thresholds

- **`t_flag ≈ 3 days`** — vacancy becomes publicly visible to the whole node (not just the vacant role's immediate network). This is a **public-goods pressure signal** — converts private economic pressure into social pressure. Players who aren't candidates for the role still feel it (e.g. a Baker stockpiling flour because Courier deliveries may slow).
- **`t_hard ≈ 14 days`** — hard backstop. If nobody has voluntarily filled the role by this point, the system guarantees continuity via forced player placement or NPC fallback (§2.5). This bounds the pressure — the system will never actually collapse, which is deliberate (49-51 philosophy: pressure that bites but doesn't compound).

2.4 Validated simulation results — encode as regression tests

At `N=50, R=3, p_m=0.20`: voluntary fills outnumber backstop fires (ratio ~1.2:1 at N=50, rising to ~2.8:1 at N=80). Starved fraction stays near 1–2% of the year. Worst-case gaps fall under 15 days at the 90th percentile. **This ratio range (1.2:1 to 2.8:1) is the target signature of a healthy system** — if a build change pushes voluntary fills below this range, the backstop is firing too often and the pressure curve needs retuning.

2.5 NPC fallback agent

When the hard backstop fires and no player fills the role, an NPC agent takes the slot with **flat, non-competitive pricing** — no negotiation, no strategic behaviour. Simulation confirmed this mildly stabilizes system volatility (~4% reduction) without eliminating competitive dynamics or dominant-player market share — it's economic content (a real, if boring, market participant), not a dead patch that pauses the game.

2.6 "Shift Cover" — the corrected offline-absence mechanic

Important correction to get right on first build — an earlier external draft got this wrong in a way that would have undermined the safety architecture in Phase 5. The offline shop-cover mechanic (what happens to a role-holder's shop while they're logged off but not vacant) must be **purely mechanical and bounded**:

- Executes **only pre-set prices the player defined in advance** while they were online.
- **Zero negotiation capability. Zero conversational capability.** It is a transactional placeholder, not an agent.
- Structurally identical in category to the NPC fallback in §2.5 — a dumb, bounded, non-agentic stand-in — just applied to offline-absence rather than confirmed vacancy.
- Do **not** build anything resembling an AI surrogate that conducts interrogations, negotiations, or free-form conversation on a player's behalf while they're offline. This was explicitly rejected during design — it has real consent and deception implications and has no place in this system regardless of how it might be framed later.

PHASE 3 — Communication Layer (The Wall, Envelopes, Rumour Mill)

Why third: this is what makes the vacancy pressure and economic tension *socially legible* — and it's also your minimum-viable-prototype gate (see §8). Two bakers plus a working rumour mill is the smallest playable slice of NODE; this phase is what makes that slice real.

3.1 Shared grammar constraint — City Wall (public) and Envelopes (private, asynchronous)

Both channels use the **identical closed-language grammar**. This is a hard constraint on message composition, not moderation after the fact — the goal is to make harassment vocabulary structurally unavailable, not to catch it after it's typed.

- **First-person only.** Every message is a statement about the sender's own state. Valid: "I feel isolated." "I feel manipulated." "I don't trust the people I sell to right now." Invalid: any sentence with a second- or third-person subject.
- **Present tense only.** No narrating past events in detail, no future threats.
- **Zero third-party reference.** No naming another player, no role-level reference (in a 50-person node, role level is functionally person-level — "the Baker" narrows a 3-person slot to near-certainty), no indirect identification ("the person who...").

Implementation approach: this should be enforced as a **constrained input** (e.g. a curated preset/template picker, or a tightly-scoped generative constraint if free text is allowed at all) rather than free-text-plus-a-profanity-filter. The safety property depends on the vocabulary being genuinely unavailable, not merely discouraged.

- Attribution is by **username only, never by role** — whether a given username holds the Baker slot is knowledge other players must earn through actual social presence, not something the engine surfaces.

- Envelopes are private (left at a player's door, visible to exist but not to read until opened) but carry the **exact same grammar constraint** as the public Wall — privacy must never become a harassment loophole. The only functional difference between the two channels is audience size, not permitted content.

3.2 Rumour mill

Not fully specified yet — this is explicitly one of the two components of the minimum viable prototype (§8) and should be built collaboratively/iteratively rather than from a rigid spec. Starting shape: secondhand, imperfect propagation of Wall/Envelope content and ambient-layer events (see Phase 4) between connected players — accounts should be allowed to be **reliably imperfect, sometimes distorted**, consistent with real gossip-transmission research, rather than perfectly faithful relay. Treat this as the component most likely to need hands-on iteration once playable — build it in a way that's easy to retune (propagation probability, distortion rate, decay) without a full re-architecture.

PHASE 4 — Identity, Camera & the Ambient Visual System

Why fourth: this is what makes Phases 1–3 *visible* and *legible* without being explicit — it's the rendering/perception layer, and it depends on all three prior systems having real state to visualize.

4.1 Camera model — one continuous space, three zoom states

Never a UI-mode teleport — zooming is a smooth transition through one coherent 3D space (isometric, RimWorld/Dwarf-Fortress-style readability at distance).

State	Shows	Hears	Knows
Zoomed In (conversation)	Immediate conversation partner(s), tight framing	Only this conversation	Full content of this exchange
Zoomed Out (city view)	Whole node from above, players as role-icons/silhouettes, colour-washed districts	Ambient only, nothing intelligible	Only what colour, movement, own connections reveal
City Wall (public layer)	Shared public message board, overlaid on/accessible from zoomed-out view	N/A (async, text)	Only first-person self-statements per §3.1

The player is never omniscient — zoomed-out is explicitly *their* sphere of influence overlaid on the city, not a god's-eye view. Zooming in should audibly/visually "seal" a conversation (ambient noise drops, confirms nobody else can overhear). Zooming out restores ambient noise but never intelligible speech from conversations that aren't the player's own.

4.2 Identity resolution — "fog of recognition"

- **Strangers:** generic silhouette + role icon only. No name, no face, no avatar detail.
- **Known connections:** full designed avatar (face, clothing, name) once a real relationship exists (direct interaction, or introduction by someone already known).
- Watching two people interact does **not** grant their identities — a player may see "a baker talking to a courier" at a distance indefinitely with no further resolution.
- **Open question, unresolved — pick one for first build and flag it as reversible:** binary resolution (silhouette flips to avatar the instant a threshold is crossed) vs. gradual resolution (dissolve/sharpen effect over repeated exposure). Gradual is likely more organic and pairs with the ripple system below; binary is cheaper and clearer. Recommend building binary first for the MVP (cheaper, testable) and treating gradual as a post-prototype polish pass.

- Avatars are fixed at creation (face, base clothing) — minor cosmetic changes allowed after, but base identity should read as a stable signature even at distance once known. This is deliberate: identity, once earned, should stay reliable.

4.3 Connection visualisation — no persistent lines

Never render a literal social graph — fifty overlapping relationship lines is illegible and also leaks information that shouldn't be available. A player only ever sees **their own** sphere of influence.

- No static drawn edges. Connections imply through **proximity/transit events** (a transient pulse where two connected points' activity overlaps) and **clustering/gravity** (frequently-interacting players visually drift closer together over time in the zoomed-out view).
- Visual reference points: RTS fog-of-war soft/organic falloff, cellular-automaton-style organic clustering (not rigid grid), weather/heat-map visualisation rather than network-graph visualisation.

4.4 Ripple layer — information propagation made visible

- A conversation generates a **ripple** at its location — a moving, decaying wave, not a static icon. Spreads outward, fades in both opacity and speed as it travels, so distance from source is always legible.

- Ripples from different origins can cross and merge (implies cross-pollination between social clusters) without revealing content or identity.
- A player can watch a ripple arrive at their own position before knowing what caused it — deliberate anticipation hook.
- **Open question:** should decay rate vary by information "weight" (a big rumour travels further/decays slower than idle chatter)? Not resolved — flag for a design pass once the base ripple system is running and you can feel whether uniform decay is enough.

4.5 Ambient mood/activity colour system — replaces day/night cycle

Two-channel model, **layered, not blended**:

Channel	Represents	Property
Hue	Type of activity	e.g. gold = trade, green = casual conversation, red/amber = tension/conflict
Saturation/glow intensity	Volume of activity	Low ambient hum by default; flares only where genuinely active

A location with brisk trade AND heavy gossip should show gold shot through with green (layered light sources), not a muddy blend (paint mixing) — implement as additive/layered rendering, not colour averaging.

Overwhelm constraint — explicit brief, treat as a hard requirement, not a suggestion: intuitive, not overwhelming. Baseline should be low, comfortable ambient hum — quiet, not blank. Only a minority of locations should be visually "loud" at any time; a bright zone should read as genuinely notable. Avoid full-saturation colour across large map areas.

Two-tier visibility (this is the central asymmetry of the whole visual system):

- **Tier 1 (everyone sees):** raw intensity/hue — "this district is hot right now" — visible regardless of connection.
- **Tier 2 (connection-gated):** *why* it's hot — actual content, players involved, reason — locked behind real relationship depth per §4.2.
- The city always tells a player something is happening; it never tells them what, unless earned socially.

Time model: no day/night reset of the ambient layer. An arbitrary daily tick exists only for housekeeping (vacancy timers, task logic — see Phase 2) and is fully decoupled from the visual/ambient layer, which stays purely activity-driven. The city should never look "asleep" just because it's 3am somewhere — it looks calm because nothing is actually happening there right now, which is a truer signal.

4.6 Layer stack (back to front, for renderer architecture)

1. Base city geometry (buildings, districts, fixed isometric camera angle)
2. Ambient colour wash (§4.5)
3. Ripple layer (§4.4)
4. Connection/proximity nodes (§4.3, player's own sphere only)
5. Player figures (§4.2, silhouette+icon or full avatar per resolution state)

Each layer should be independently legible — you should be able to toggle any one off during dev and the others should still make visual sense.

4.7 Relationship-gated interaction resolution

A transaction between strangers reads as a flat, functional trade-coloured event to any distant observer. The same transaction between players with real relationship depth can carry a **second, sharper signal layered underneath** (e.g. a personal-money undertone if pricing deviates from market rate) — visible only to the parties involved (and readable by them precisely because they know each other, not because the game announces it). This is what makes betrayal/exploitation possible: it requires the relationship investment to exist first, which is the actual cost of access to exploit it.

PHASE 5 — Voice & Safety Architecture

Why fifth, and why it's scaffolding not a finished system: live voice is the one channel that can't be grammar-constrained the way Phase 3 is, so it needs active moderation rather than structural prevention. Build the detection/flagging/review *pipeline architecture* now so the system has somewhere to plug in real moderation later — but do **not** treat any specific retention period, consent flow, or enforcement policy below as final. These carry real legal weight (GDPR, UK Online Safety Act) and must be reviewed by a lawyer with gaming/data-protection experience before this goes anywhere near real users with real voice data.

5.1 Two independent detection triggers, one review queue

- **(a) Live automated detection (background, passive):** voice analysed in real time for harassment/hate speech/abuse as it happens, not stored wholesale and reviewed after the fact. Only flagged moments retained as short clips with timestamp/context for human review — no permanent recording of every conversation is required or desired.
- **(b) Player-initiated flagging (active):** any player can flag an interaction in the moment, submitting the recent exchange for priority review. Needs a light cooldown/pattern-check on repeat flaggers so the flag mechanism itself can't be weaponised as a harassment tool.

Both feed the same review queue — human-in-the-loop, automated systems surface high-confidence events, a person makes the final call.

5.2 Retention & disclosure — build the architecture to support these; do not hardcode final values

- No blanket recording — only flagged segments retained.
- Retention window: long enough for review/appeal, not indefinite. **[LEGAL REVIEW REQUIRED before shipping a real number.]**
- Clear upfront disclosure at signup (not buried in ToS): voice is monitored for safety, flagged content may be human-reviewed, never sold/shared with third parties.
- **GDPR posture:** voice recordings are personal data; consent must be freely given, informed, specific. **[LEGAL REVIEW REQUIRED — lawful basis, likely legitimate interest for safety purposes, needs confirmation against GDPR Art. 6, not a design-level decision.]**

5.3 Enforcement

Graduated ladder — warning, temporary voice-mute, suspension, ban — depending on severity/pattern. Players told when action is taken and at minimum the general category of why.

5.4 Summary table

Channel	Constraint type	Harassment prevention mechanism
City Wall	Structural (grammar)	Vocabulary unavailable — see Phase 3
Envelopes	Structural (grammar)	Same grammar as Wall, any audience size
Voice	Active (moderation)	Live detection + flagging → human review → graduated enforcement

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PHASE 6 — Stress-Testing & Balance Harness

Why last, but build hooks for it from Phase 1 onward: everything above is a hypothesis until it's been run against real or simulated player behaviour. You should have a headless simulation harness (started in §1.5) that can, by the end of this phase:

- Sweep `N`, `R`, `γ`, per-role headcounts and reproduce the stability curves in §1.4.
- Sweep vacancy parameters (`β`, `T_pain`, `v_boost`, `t_flag`, `t_hard`) and reproduce the voluntary:backstop ratio range in §2.4.
- Flag automatically if any role slot spends meaningful time at exactly `n=2` (the known instability cliff).

- Log enough telemetry once real players are involved to eventually calibrate the **[CALIBRATED — provisional]** parameters against real behaviour rather than today's guesses.

This phase should never really "finish" — treat it as a permanent regression/tuning tool that gets used every time a mechanic changes, not a one-time build task.

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7. Explicitly Unresolved — Do Not Silently Resolve These

Flag rather than guess on any of the following. They were identified as open in design and deliberately left that way:

- **Ruin Floor (`R(t)`):** an economic floor invariant intended to prevent indefinite ruin for a player — "a recovery path a determined attacker cannot fully seal." Never mathematically specified. This was flagged as one of the two most likely points of real-world failure (alongside density/population numbers), discoverable only with live players. Do not invent a specific formula for this without design sign-off.
- **Density numbers** (players per district, suppliers per good) — likely only discoverable through actual playtesting, not derivable in advance.
- Binary vs. gradual identity resolution (§4.2) — build binary first, flag as reversible.

- Exact colour palette per activity type (§4.5) — only rough examples exist (gold/green/red-amber).
- Ripple decay-rate variance by information "weight" (§4.4).
- How the City Wall visually integrates with the ambient colour system — own colour signature, or separate UI plane? Not decided.
- All of §5.2's legal specifics.

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8. Minimum Viable Prototype — The Actual First Milestone

Two Bakers plus a working rumour mill. This is deliberately small: it requires Phase 1 (at minimum, the Baker/Bertrand side, with a placeholder or hardcoded flour price rather than a full Miller layer if that's faster to stand up first), a slice of Phase 3 (Wall + rumour propagation), and enough of Phase 4 to make the interactions visually legible. It does **not** require Phase 2's full vacancy math, Phase 5's voice system, or Phase 6's full harness on day one — those can follow once the core loop is provably fun and legible with the smallest possible slice of the system running.

Growth strategy beyond the prototype: phased, social-media-driven recruitment matched tightly to role-slot openings — start with one Baker for a small founding population, open slots only as waitlist demand confirms fill capacity, let real churn data find the stable band rather than targeting a fixed headcount from day one.

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9. How to Work With This Document

- Everything marked **[CALIBRATED — provisional]** is expected to change once real data exists. Don't treat it as sacred.
- Everything marked as an open question in §7 should be flagged back for a design decision, not silently resolved with a plausible-sounding default.
- The regression-test-worthy findings in §1.4 and §2.4 are the closest thing to "hard truth" in this document — they came from direct simulation, not just design reasoning. Preserve them as automated tests so future refactors can't quietly break the stability properties that were actually proven.
- This is explicitly meant to be built, played, and argued with — structural rules exist, but they're subject to revision by what actually happens once people are playing. Say so plainly whenever something built doesn't hold up in practice.